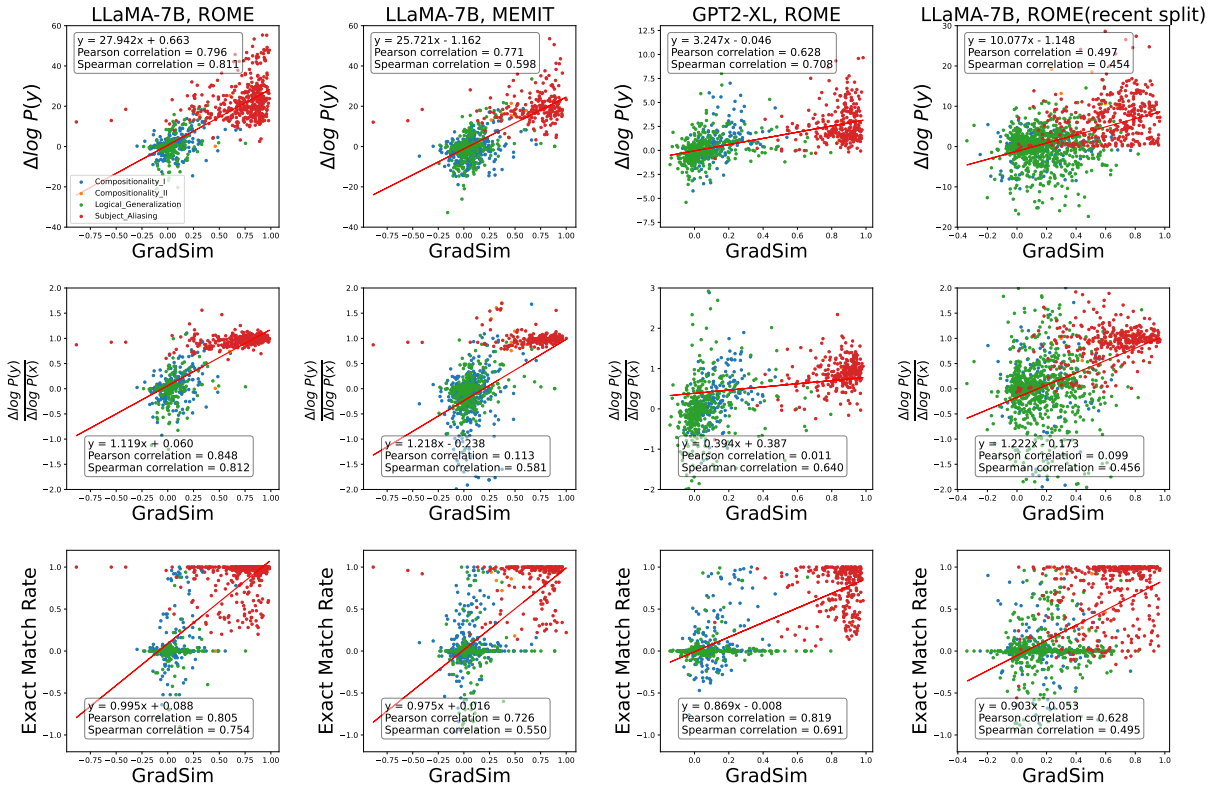



Figure 6: Main results of evaluating the correlation between ripple effect performances and GradSim values labeled with different task names

additional experiments, we only compute on GPT-XL due to computational resource limitations. The experiments are done with the popular subset in RippleEdits. Here are the experiment results:

| Method | Spearman Correlation | Pearson Correlation | Best Fit Line |
|---|---|---|---|
| MEND | 0.628 | 0.501 | $y = 0.845x - 0.177$ |
| FT | 0.231 | 0.731 | $y = 0.430x + 0.119$ |

Table 1: Main experiments with other knowledge editing method

These experimental results demonstrate the versatility of the GradSim indicator.

### B.2 Experiments on other dataset

We conducted experiments on all data in RippleEdits to demonstrate the indicator's effectiveness. Additional experiments are also done on the 2-hop subset of MQUAKE dataset (Zhong et al., 2023b) to support the validity of our result. The results are as follows

| Method | Model | Spearman Correlation | Pearson Correlation | Best Fit Line |
|---|---|---|---|---|
| ROME | GPT2-XL | 0.767 | 0.783 | $y = 1.164x - 8.923$ |

Table 2: Results on MQUAKE

## C Extra Results of GradSim

The RippleEdits dataset comprises six distinct tasks: Logical Generalization (LG), Compositionality I (CI), Compositionality II (CII), Subject Aliasing (SA), Preservation (PV), and Relation Specificity (RS) (Cohen et al., 2023). These tasks are designed to evaluate different aspects of knowledge editing in neural networks. Specifically, LG, CI, CII, and SA are tasks where the model is expected to manifest new knowledge in response to edits made to existing entries. Conversely, in the RS and PV tasks, the existing knowledge should remain unaltered post-editing, as these tasks are designed to test the model's ability to preserve information that is logically independent of the changes applied.

In Figure 2, we analyze data from these tasks to illustrate a positive correlation between ripple effect performance and GradSim values across the four tasks explicitly associated with knowledge updates, hereafter referred to as *ripple tasks*. Each data point in Figure 6 is labeled to demonstrate the consistent applicability of the GradSim metric across the individual sub-tasks. In contrast, Figure 7 focuses on the two *non-ripple tasks* (RS and PV), where no significant correlation is observed between GradSim values and $\Delta \log P(y)$.

The comparison underscores that GradSim is a critical metric for evaluating ripple effects, as it shows no significant impact in the *non-ripple tasks*, confirming its relevance specifically in contexts where knowledge modifications are expected.

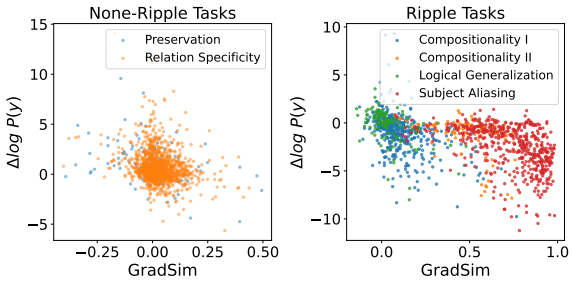



Figure 7: Comparison of the Correlation on None-Ripple Tasks and Ripple Tasks.

## D How to Represent Knowledge Distribution?

In this paper, we utilize gradients to represent the distribution of knowledge. To support this approach, we conducted preliminary experiments that lend credence to the underlying rationale of this intuition.

### D.1 Does the way that we express a piece of knowledge change the knowledge distribution?

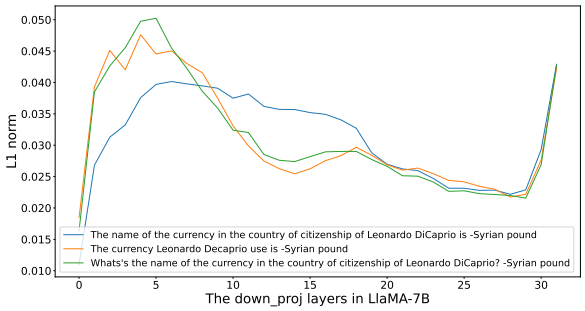



Figure 8: L1 Norm Distribution over LlaMA-7B

In this study, we calculate the gradient of a specific piece of knowledge, "The name of the currency in the country of citizenship of Leonardo DiCaprio is the Syrian pound," along with its variants: "The currency Leonardo DiCaprio uses is the Syrian pound" and "What's the name of the currency in the country of citizenship of Leonardo DiCaprio? Syrian pound." We then plot the L1 norm of the gradient across the 32 downward projection layers of LlaMA7b. Prior research suggests that these layers are particularly adept at storing knowledge. Our results indicate that the distribution across these variants is remarkably consistent, suggesting that a piece of knowledge may be encoded within specific parameters of a large language model.